VALIDATION REPORT
OPERATIONAL QUALIFICATION
CORRELATION ANALYSIS MODULE

JR Validated Environment — Correlation Analysis Module v1.0

Field	Value
Document Number	JR-VR-CORR-001
Title	Validation Report — Operational Qualification, Correlation Analysis Module
Validation Plan	JR-VP-CORR-001 v1.0
System	JR Validated Environment — Correlation Analysis Module
Module Version	1.0
Document Version	1.0
Execution Date	2026-03-20
OQ Result	PASS — 45 / 45 tests passed, 0 failures, 0 deviations
Author	Joep Rous
Reviewer	[Name]
Approver	[Name]

Version History

Version	Date	Author	Description
1.0	2026-03-20	Joep Rous	Initial OQ execution report. All 45 test cases passed. No deviations.

Approval Signatures

Role	Name	Signature	Date
Author	Joep Rous		2026-03-20
Reviewer	[Name]		
Approver (QA)	[Name]		

1. Purpose

This document reports the execution and outcome of the Operational Qualification (OQ) test suite for the JR Validated Environment Correlation Analysis Module, version 1.0. It provides objective evidence that all Corr scripts perform as specified in the OQ Validation Plan JR-VP-CORR-001 v1.0 under the defined test conditions.

This report records the test environment, execution results for all 45 test cases, and the formal OQ conclusion.

2. Scope

2.1 In Scope

- All 4 R scripts in repos/corr/R/ (jrc_corr_pearson, jrc_corr_spearman, jrc_corr_regression, jrc_corr_passing_bablok)
- Correct computation of correlation coefficients, regression coefficients, CI bounds, and p-values for valid inputs
- Correct rejection of invalid inputs with informative error messages
- Bypass-protection verification (RENV_PATHS_ROOT check)
- PNG output to ~/Downloads/

2.2 Out of Scope

- Infrastructure scripts in bin/ and admin/ (covered by JR-IQ-001)
- Performance qualification (PQ) with real production data
- Validation of R interpreter and third-party packages (ggplot2, grid)
- Statistical validation of the underlying methods (Pearson, Spearman, OLS, Passing-Bablok) — method references are cited in JR-VP-CORR-001 Section 3

3. Reference Documents

Document ID	Title	Location
JR-VP-CORR-001 v1.0	Validation Plan — OQ, Correlation Analysis Module	repos/corr/docs/corr_validation_plan.pdf
JR-VP-002 v1.0	Validation Plan — OQ, Community Script Suite v1.1.0	docs/statistics_validation_plan.pdf
JR-IQ-001	IQ Execution Evidence	docs/core_IQ_validation_plan.pdf
OQ Evidence	Corr OQ Execution Evidence (pytest output + header)	~/.jrscript/MyProject/validation/corr_oq_execution_20260320T225218.txt

4. Test Environment

Item	Value
Execution date / time	2026-03-20T22:52:18

Completion date / time	2026-03-20T22:52:43
Host	Joeps-MacBook-Air.local
Operating system	macOS 15.7.3 (Darwin 24.6.0, aarch64)
R version	4.5.0
Python version	3.11.9
pytest version	8.3.4
OQ virtual environment	~/.venvs/MyProject_oq/ (reused from core OQ suite)
Script entry point	bin/jrrun (sets RENV_PATHS_ROOT, loads renv library)
Test runner	repos/corr/admin_corr_oq
Test directory	repos/corr/oq/
Test data directory	repos/corr/oq/data/ (6 committed synthetic CSV files)
Evidence file	~/.jrscript/MyProject/validation/corr_oq_execution_20260320T225218.txt

5. Test Data

All test data files are synthetic CSV files committed to repos/corr/oq/data/ and included in the project integrity check (admin/project_integrity.sha256). No real patient or production data is used.

File	Content	Used by
corr_linear.csv	n=10, y≈2x+1 with small noise; r≈0.999 strong positive correlation	TC-CORR-P-001..005, TC-CORR-S-001..005, TC-CORR-R-001..005
corr_negative.csv	n=10, y≈−2x+22; r≈−0.999 strong negative correlation	TC-CORR-P-010, TC-CORR-S-010
corr_uncorrelated.csv	n=10, irregular y values; |r| < 0.5, not significant	Supplementary
corr_method_comp.csv	n=9, y≈1.1x+2 with small noise; method comparison data	TC-CORR-PB-001..006, TC-CORR-PB-011
corr_small.csv	n=2 pairs only — triggers minimum-n error	TC-CORR-P-008, TC-CORR-S-008, TC-CORR-R-008, TC-CORR-PB-009
corr_nonnumeric.csv	y column contains letters A, B, C — triggers non-numeric error	TC-CORR-P-009, TC-CORR-S-009, TC-CORR-R-009, TC-CORR-PB-010

6. Test Execution Results

6.1 Summary

Item	Value
Total test cases	45
Passed	45
Failed	0
Errors	0
Duration	25.13 s
OQ result	PASS

6.2 Results by Test File

Test file	Script covered	Tests	Result

test_corr_passing_bablok.py	jrc_corr_passing_bablok	12	12 / 12 PASS
test_corr_pearson.py	jrc_corr_pearson	11	11 / 11 PASS
test_corr_regression.py	jrc_corr_regression	11	11 / 11 PASS
test_corr_spearman.py	jrc_corr_spearman	11	11 / 11 PASS

6.3 Individual Test Case Results

All 45 test cases passed. The full pytest output, including individual test IDs, pass/fail status, and timing, is recorded in the evidence file:

~/.jrscript/MyProject/validation/corr_oq_execution_20260320T225218.txt

That file also contains the evidence header (host, R version, Python version, pytest version, timestamp) and the OQ RESULT: PASS footer. The per-test results are reproduced in the table below for traceability.

Test Case ID	Description	Expected	Result
TC-CORR-P-001	Happy path — exit 0, 'Pearson' in output	Exit 0, 'Pearson' and 'Pearson r:' present	PASS
TC-CORR-P-002	Pearson r label present	Exit 0, 'Pearson r:' in output	PASS
TC-CORR-P-003	CI values present in output	Exit 0, '[' and ']' in output	PASS
TC-CORR-P-004	p-value present in output	Exit 0, 'p-value' in output	PASS
TC-CORR-P-005	PNG written to ~/Downloads/	PNG present, recent mtime	PASS
TC-CORR-P-006	No arguments → usage message	Exit ≠ 0, 'Usage' in output	PASS
TC-CORR-P-007	File not found	Exit ≠ 0, 'not found' in output	PASS
TC-CORR-P-008	n < 3 (corr_small.csv) → error	Exit ≠ 0, '3' in output	PASS
TC-CORR-P-009	Non-numeric column → error	Exit ≠ 0	PASS
TC-CORR-P-010	Negative correlation → r < 0 in output	Exit 0, '-0.' in output	PASS
TC-CORR-P-011	Bypass protection — direct Rscript call	Exit ≠ 0, 'RENV_PATHS_ROOT' in output	PASS
TC-CORR-S-001	Happy path — exit 0, 'Spearman' in output	Exit 0, 'Spearman' present	PASS
TC-CORR-S-002	Spearman rho label present	Exit 0, 'Spearman rho:' in output	PASS
TC-CORR-S-003	p-value present in output	Exit 0, 'p-value' in output	PASS
TC-CORR-S-004	CI note present	Exit 0, 'Confidence interval' note present	PASS
TC-CORR-S-005	PNG written to ~/Downloads/	PNG present, recent mtime	PASS
TC-CORR-S-006	No arguments → usage message	Exit ≠ 0, 'Usage' in output	PASS
TC-CORR-S-007	File not found	Exit ≠ 0, 'not found' in output	PASS
TC-CORR-S-008	n < 3 → error	Exit ≠ 0	PASS
TC-CORR-S-009	Non-numeric column → error	Exit ≠ 0	PASS
TC-CORR-S-010	Negative correlation → rho < 0 in output	Exit 0, '-0.' in output	PASS
TC-CORR-S-011	Bypass protection — direct Rscript call	Exit ≠ 0, 'RENV_PATHS_ROOT' in output	PASS
TC-CORR-R-001	Happy path — exit 0, 'Regression' in output	Exit 0, 'Regression' present	PASS
TC-CORR-R-002	Slope label present	Exit 0, 'Slope' in output	PASS
TC-CORR-R-003	Intercept label present	Exit 0, 'Intercept' in output	PASS
TC-CORR-R-004	R-squared present in output	Exit 0, 'R-squared' in output	PASS
TC-CORR-R-005	PNG written to ~/Downloads/	PNG present, recent mtime	PASS
TC-CORR-R-006	No arguments → usage message	Exit ≠ 0, 'Usage' in output	PASS
TC-CORR-R-007	File not found	Exit ≠ 0, 'not found' in output	PASS
TC-CORR-R-008	n < 3 → error	Exit ≠ 0	PASS

TC-CORR-R-009	Non-numeric column → error	Exit ≠ 0	PASS
TC-CORR-R-010	Slope ≈ 2.0 for corr_linear.csv	Exit 0; '1.9' or '2.0' in output	PASS
TC-CORR-R-011	Bypass protection — direct Rscript call	Exit ≠ 0, 'RENV_PATHS_ROOT' in output	PASS
TC-CORR-PB-001	Happy path — exit 0, 'Passing-Bablok' in output	Exit 0, 'Passing-Bablok' present	PASS
TC-CORR-PB-002	Slope label present	Exit 0, 'Slope' in output	PASS
TC-CORR-PB-003	Intercept label present	Exit 0, 'Intercept' in output	PASS
TC-CORR-PB-004	Cusum test result present	Exit 0, 'Cusum' in output	PASS
TC-CORR-PB-005	Proportionality test present	Exit 0, slope=1 test in output	PASS
TC-CORR-PB-006	PNG written to ~/Downloads/	PNG present, recent mtime	PASS
TC-CORR-PB-007	No arguments → usage message	Exit ≠ 0, 'Usage' in output	PASS
TC-CORR-PB-008	File not found	Exit ≠ 0, 'not found' in output	PASS
TC-CORR-PB-009	n < 3 → error	Exit ≠ 0	PASS
TC-CORR-PB-010	Non-numeric column → error	Exit ≠ 0	PASS
TC-CORR-PB-011	Slope ≈ 1.1 for corr_method_comp.csv	Exit 0; '1.1' in output	PASS
TC-CORR-PB-012	Bypass protection — direct Rscript call	Exit ≠ 0, 'RENV_PATHS_ROOT' in output	PASS

7. Deviations from OQ Plan (JR-VP-CORR-001 v1.0)

No deviations from the approved OQ plan were identified during this OQ execution.

ID	Description	Resolution	Status
—	None	—	N/A

8. Requirements Traceability Matrix

The table below maps each User Requirement from JR-VP-CORR-001 to the test cases executed and their result.

UR	Requirement (summary)	Test Cases	Result
UR-CORR-001	Pearson r, CI (Fisher z), t-stat, p-value	TC-CORR-P-001..004	4 / 4 PASS
UR-CORR-002	Pearson correlation classification (weak/moderate/strong, direction)	TC-CORR-P-001, TC-CORR-P-010	2 / 2 PASS
UR-CORR-003	Pearson scatter PNG to ~/Downloads/	TC-CORR-P-005	1 / 1 PASS
UR-CORR-004	Spearman rho, S statistic, p-value	TC-CORR-S-001..003	3 / 3 PASS
UR-CORR-005	Spearman classification; CI note displayed	TC-CORR-S-001, TC-CORR-S-004, TC-CORR-S-010	3 / 3 PASS
UR-CORR-006	Spearman rank scatter PNG to ~/Downloads/	TC-CORR-S-005	1 / 1 PASS
UR-CORR-007	OLS regression: slope, intercept, R^2, CI, F-stat, p-value	TC-CORR-R-001..004, TC-CORR-R-010	5 / 5 PASS
UR-CORR-008	Regression two-panel PNG (scatter + residuals) to ~/Downloads/	TC-CORR-R-005	1 / 1 PASS
UR-CORR-009	Passing-Bablok slope, intercept, CI (native algorithm)	TC-CORR-PB-001..003, TC-CORR-PB-011	4 / 4 PASS
UR-CORR-010	Proportionality test (slope=1) and bias test (intercept=0)	TC-CORR-PB-005	1 / 1 PASS
UR-CORR-011	Cusum linearity test	TC-CORR-PB-004	1 / 1 PASS
UR-CORR-012	Passing-Bablok PNG with identity line to ~/Downloads/	TC-CORR-PB-006	1 / 1 PASS

UR-CORR-013	--xcol / --ycol column selection	TC-CORR-P-001, TC-CORR-S-001, TC-CORR-R-001, TC-CORR-PB-001	4 / 4 PASS
UR-CORR-014	All scripts — RENV_PATHS_ROOT bypass protection	TC-CORR-P-011, TC-CORR-S-011, TC-CORR-R-011, TC-CORR-PB-012	4 / 4 PASS
UR-CORR-015	All scripts — no-arguments usage message	TC-CORR-P-006, TC-CORR-S-006, TC-CORR-R-006, TC-CORR-PB-007	4 / 4 PASS
UR-CORR-016	All scripts — file not found error	TC-CORR-P-007, TC-CORR-S-007, TC-CORR-R-007, TC-CORR-PB-008	4 / 4 PASS
UR-CORR-017	All scripts — reject n < 3	TC-CORR-P-008, TC-CORR-S-008, TC-CORR-R-008, TC-CORR-PB-009	4 / 4 PASS
UR-CORR-018	All scripts — reject non-numeric column	TC-CORR-P-009, TC-CORR-S-009, TC-CORR-R-009, TC-CORR-PB-010	4 / 4 PASS

9. OQ Conclusion

The Operational Qualification of the JR Validated Environment Correlation Analysis Module v1.0 is complete. All 45 test cases defined in JR-VP-CORR-001 were executed on 2026-03-20 and passed with no failures, no errors, and no deviations.

All 18 user requirements (UR-CORR-001 through UR-CORR-018) are covered by at least one passing test case. The OQ acceptance criteria stated in JR-VP-CORR-001 Section 10 are fully met.

The Correlation Analysis module is hereby declared OPERATIONALLY QUALIFIED and released for use in design verification activities in accordance with the JR Validated Environment quality system.

OQ Evidence file: ~/.jrscript/MyProject/validation/corr_oq_execution_20260320T225218.txt